$$\hat{q}_t = a_t, \qquad \hat{\dot{q}}_t = 0,$$

$$\tau_t = K_p(\hat{q}_t - q_t) + K_d(0 - \dot{q}_t),$$

$$K_p = 55, \qquad K_d = 0.8,$$

$q_t, \hat{q}_t, \dot{q}_t, \tau_t \in \mathbb{R}^{12}$.